



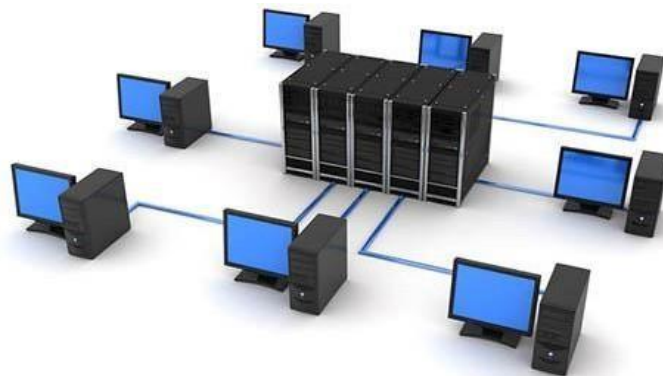
ICT 1201

Fundamentals of Computer
Systems

Computers and communication

What is a network?

- ❑ A computer network consists of two or more computers, and typically other devices as well (such as printers, external hard drives, modems and routers), that are linked together by using a transmission media, so that they can communicate with each other and there by exchange commands and share data, hardware and other resources.
- ❑ The devices on a network are referred to as nodes.

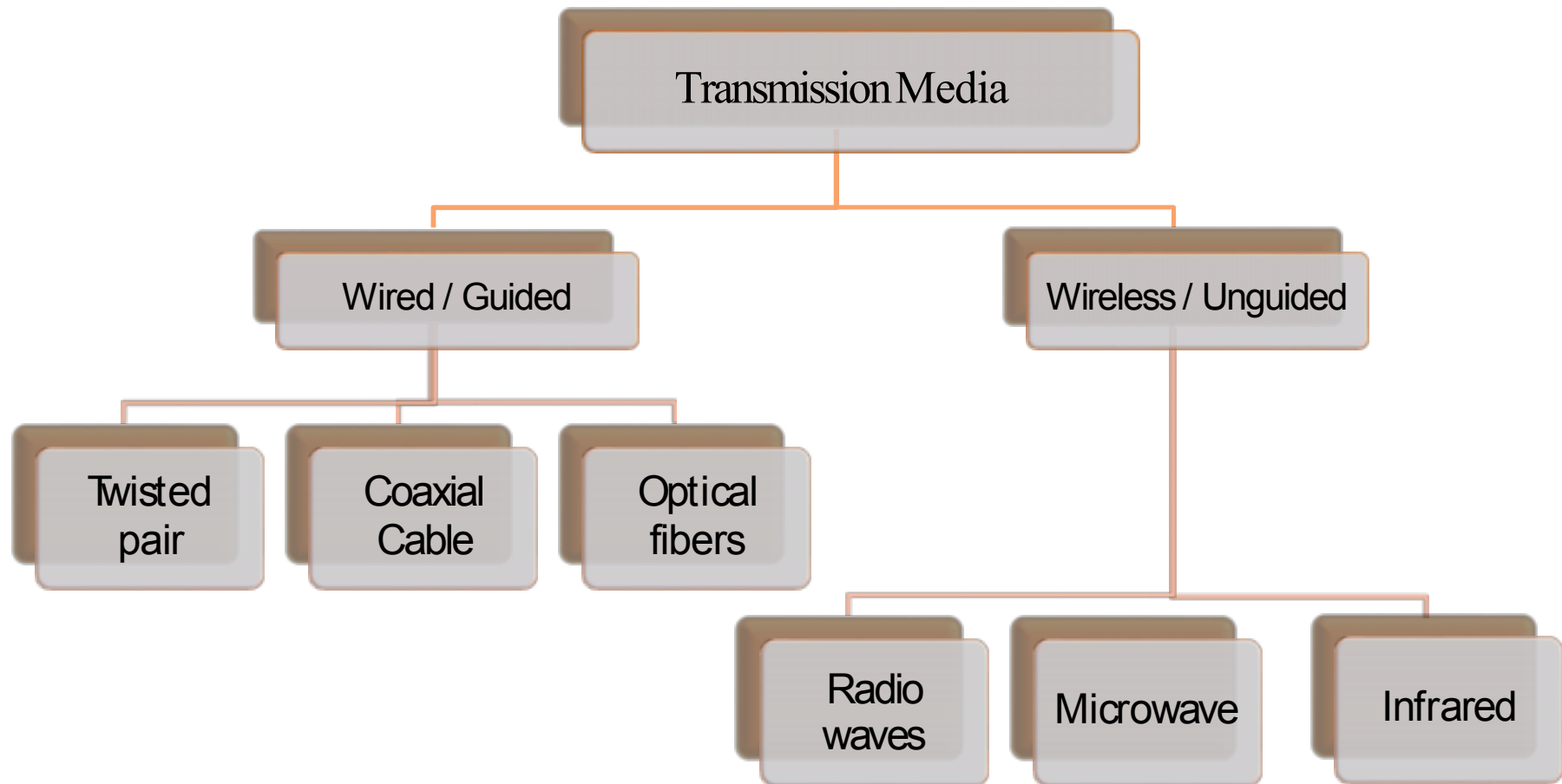


Components of network

- ❑ Computer system with network card
- ❑ Transmission media / Communication channel
- ❑ Communication software
- ❑ Networking devices
- ❑ Data / Information
- ❑ Networked / Shared resources
- ❑ Users

Transmission Media

- ❑ Is a pathway that carries the information from sender to receiver.
- ❑ Data is transmitted normally through electrical or electromagnetic signals.
 - ❑ Electrical signal in the form of current.
 - ❑ Electromagnetic signal is series of electromagnetic energy pulses at various frequencies.



Transmission Media

- ❑ Wired or Guided Media

- ❑ Are the cables that are tangible or have physical existence and are limited by the physical geography.
- ❑ E.g. co-axial cable, twisted pair cable and fiber optical cable

- ❑ Wireless or Unguided Media

- ❑ The ways of transmitting data without using any cables.
- ❑ E.g. Microwave, Radio wave, Infrared

Transmission media

- Home work
 - Fill in the blanks

Guided Media Types			
Media Type	Description	Advantages	Disadvantages
Twisted pair cable			
Coaxial cable			
Fiber optic cables			
Unguided Media Types			
Radio waves			
Microwave			
Infrared			

Transmission media

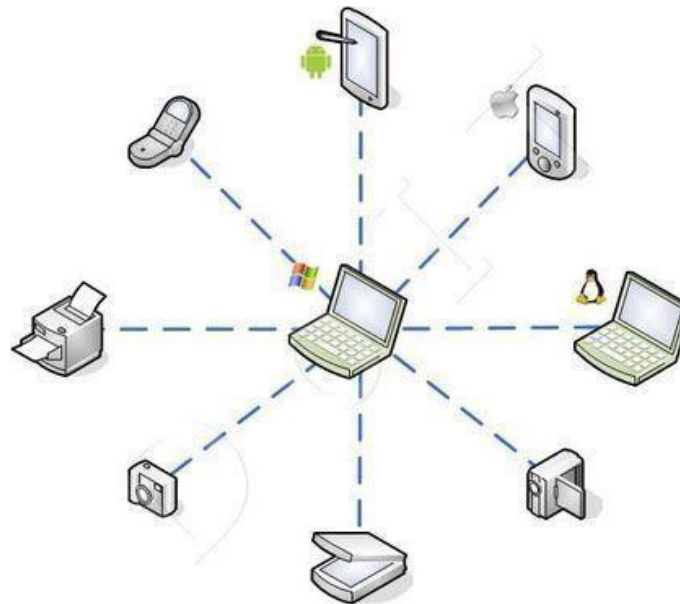
- ❑ The data transmission capabilities of various medias vary differently depending upon following factors.
 - ❑ **Bandwidth:** Data carrying capacity of a medium.
 - ❑ **Radiation:** Leakage of signal from the medium due to undesirable electrical characteristics of the medium.
 - ❑ **Noise Absorption:** Susceptibility of the media to external noise that can cause distortion of data signal.
 - ❑ **Attenuation:** The amount of energy lost depends on frequency.

Types of Networks

- ❑ Personal Area Network (PAN)
 - ❑ Is a computer network organized around an individual person within a single building.
 - ❑ This could be inside a small office or residence.
 - ❑ A typical PAN would include one or more computers, telephones, devices, video game console and other personal entertainment devices.

Types of Networks

- ❑ Personal Area Network(PAN)
 - ❑ If multiple individuals use the same network within a residence, the network is sometimes referred to as a home area network (HAN)

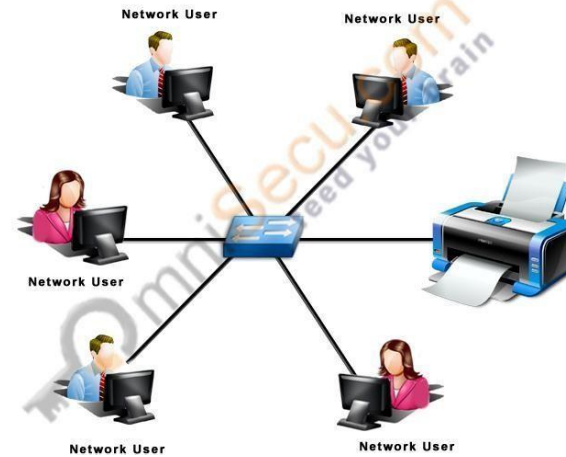
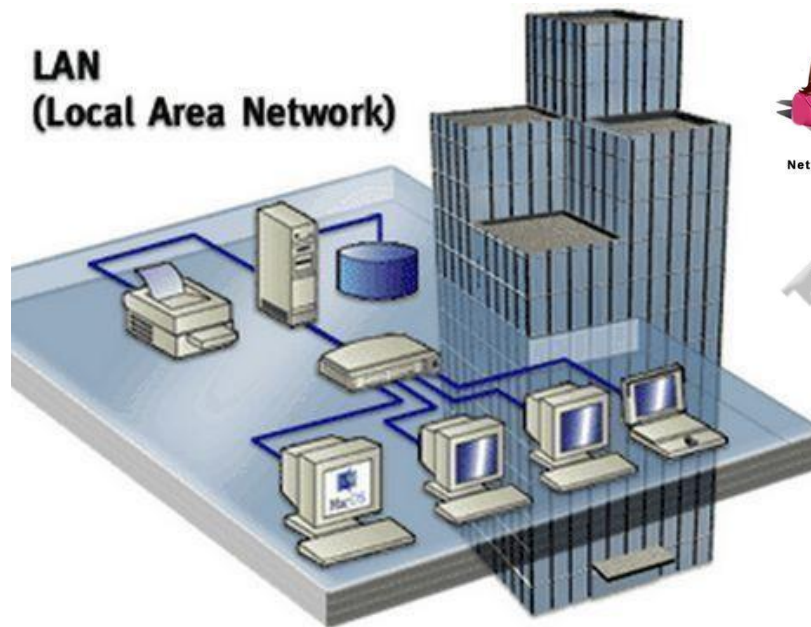


Types of Networks

- ❑ Local Area Network (LAN)
 - ❑ A network that serves users within a confined area such as a building.
 - ❑ Very useful for sharing resources, such as data storage and printers.
 - ❑ A LAN typically relies mostly on wired connections for increased speed and security.
 - ❑ If a LAN is entirely wireless, it is known as wireless local area network (WLAN)

Types of Networks

❑ Local Area Network (LAN)

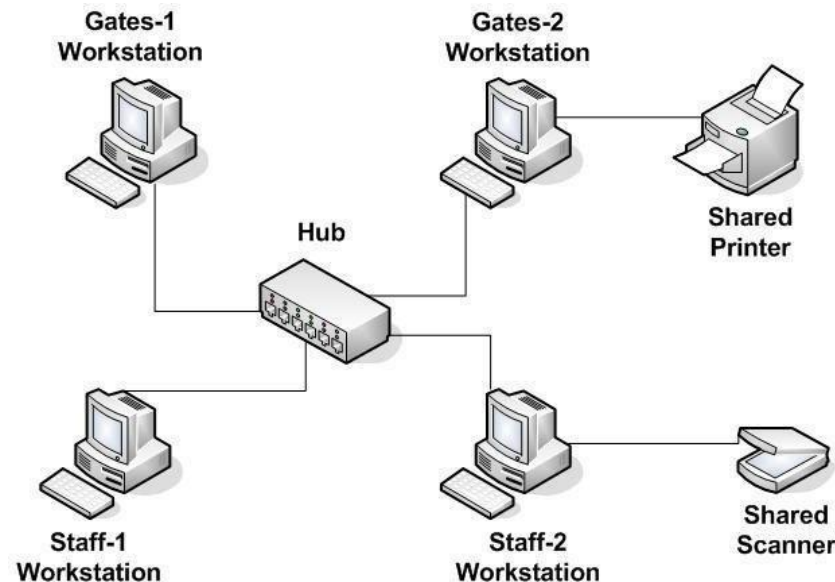


Types of Networks

- ❑ Local Area Network (LAN)

- ❑ Types of LAN

- ❑ **Peer to peer** : All computers on the network communicate and share resources directly with one another.

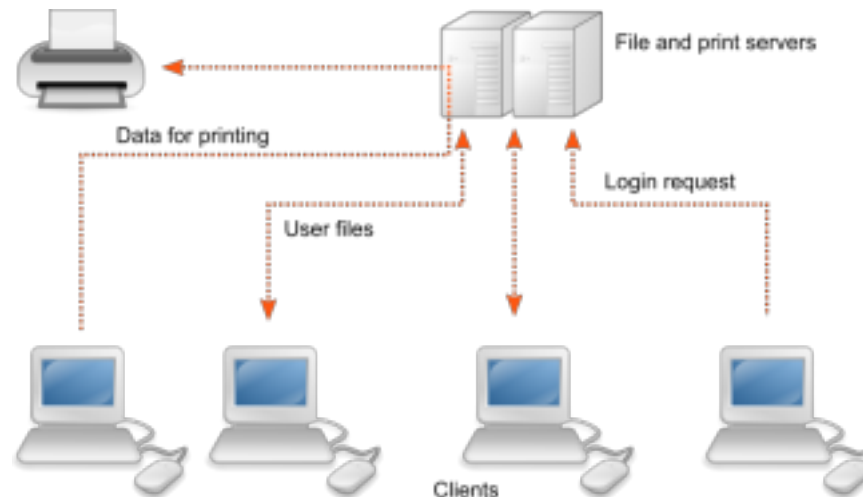


Types of Networks

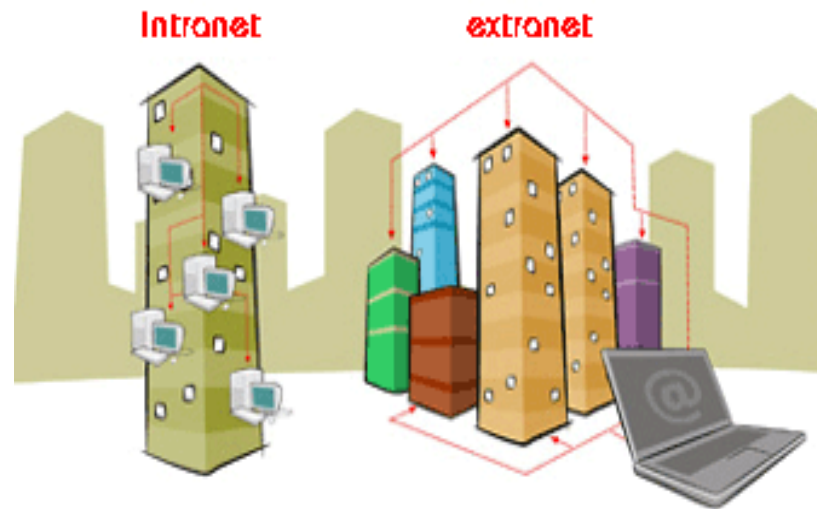
- ❑ Local Area Network (LAN)

- ❑ Types of LAN

- ❑ **Client Server** : Consist of clients – computers requesting for files or applications, and server(s) that provide applications and services to another computers (clients).

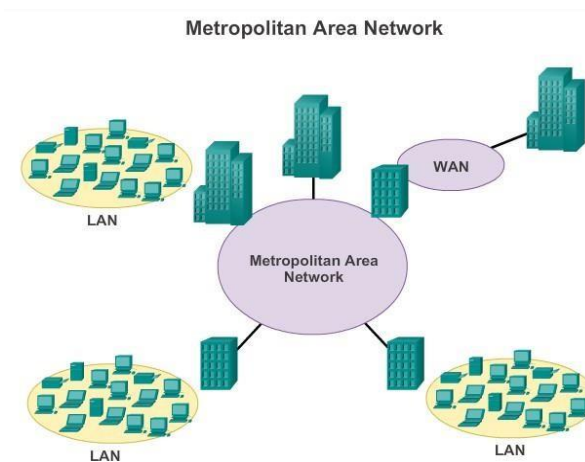


- **Intranet** : This is referred to a private network that can be accessed only by company staff to share data, share resources and communicate with each other.
- **Extranet** : A particular type of intranet connectively in which company employees connected with partners, vendors and suppliers that are authorized to use the network.



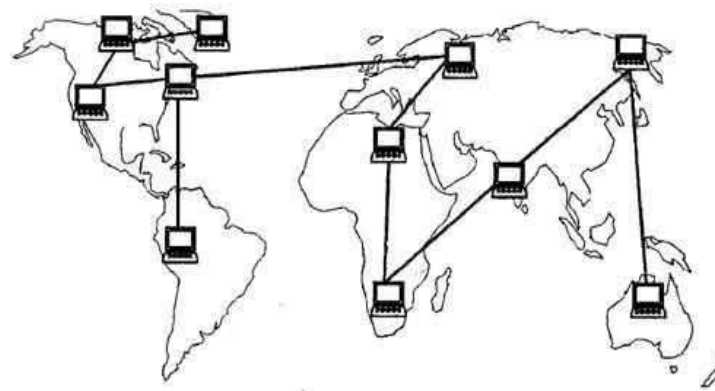
Types of Networks

- ❑ Metropolitan Area Network(MAN)
 - ❑ A computer network across an entire city, college campus or small region.
 - ❑ A MAN is often used to connect several LANs together to form a bigger network.
 - ❑ When this type of network specially designed for a college campus, it is referred to as a campus area network (CAN).



Types of Networks

- ❑ Wide Area Network(WAN)
 - ❑ Occupies a very large area, such as entire country or the entire world.
 - ❑ A WAN can contain multiple smaller networks such as LANs or MANs.
 - ❑ The Internet is the best known example of public WAN.



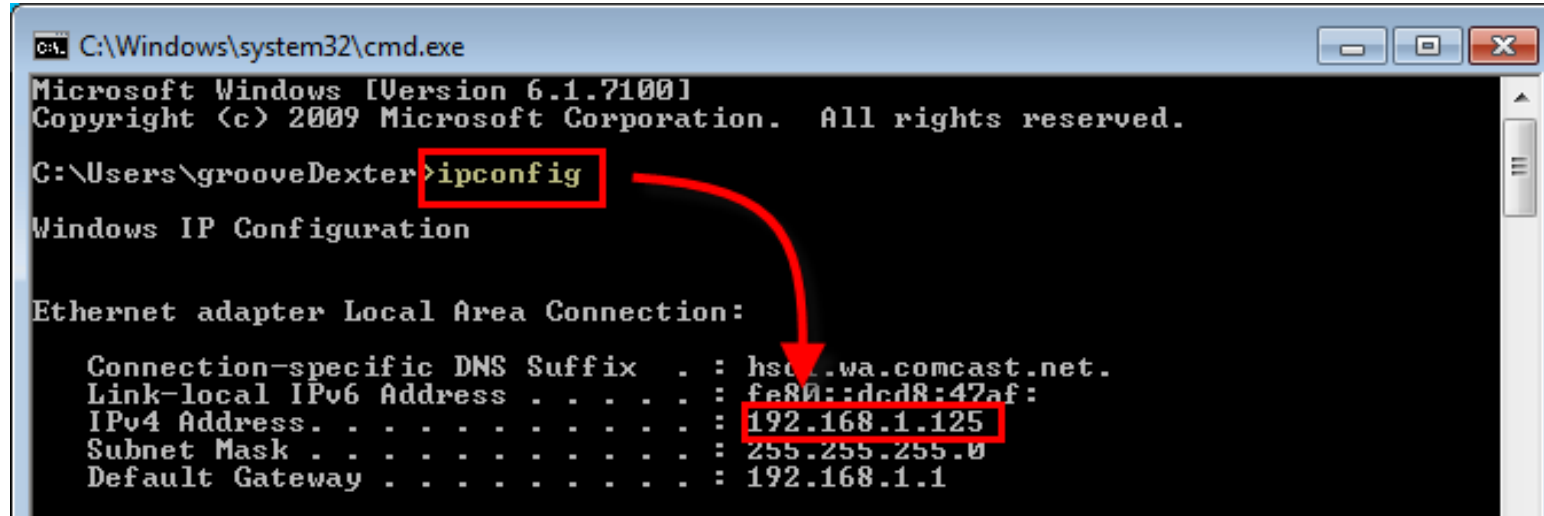
Wide area network

- ❑ **Internet** : Is a global network connecting millions of computers. It is a network of networks that's operate using a standardized set of communication protocols called TCP/IP (Transmission Control Protocol/Internet Protocol)

What is an IP address?

- ❑ Every machine on a network has a unique identifier.
- ❑ Just as you would address a letter to send in the mail, computers use the unique identifier to send data to a specific computers on a network.
- ❑ Usually it is 32 bit dotted decimal number.
 - ❑ E.g. 128.143.137.144

What is an IP address?



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7100]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\grooveDexter>ipconfig

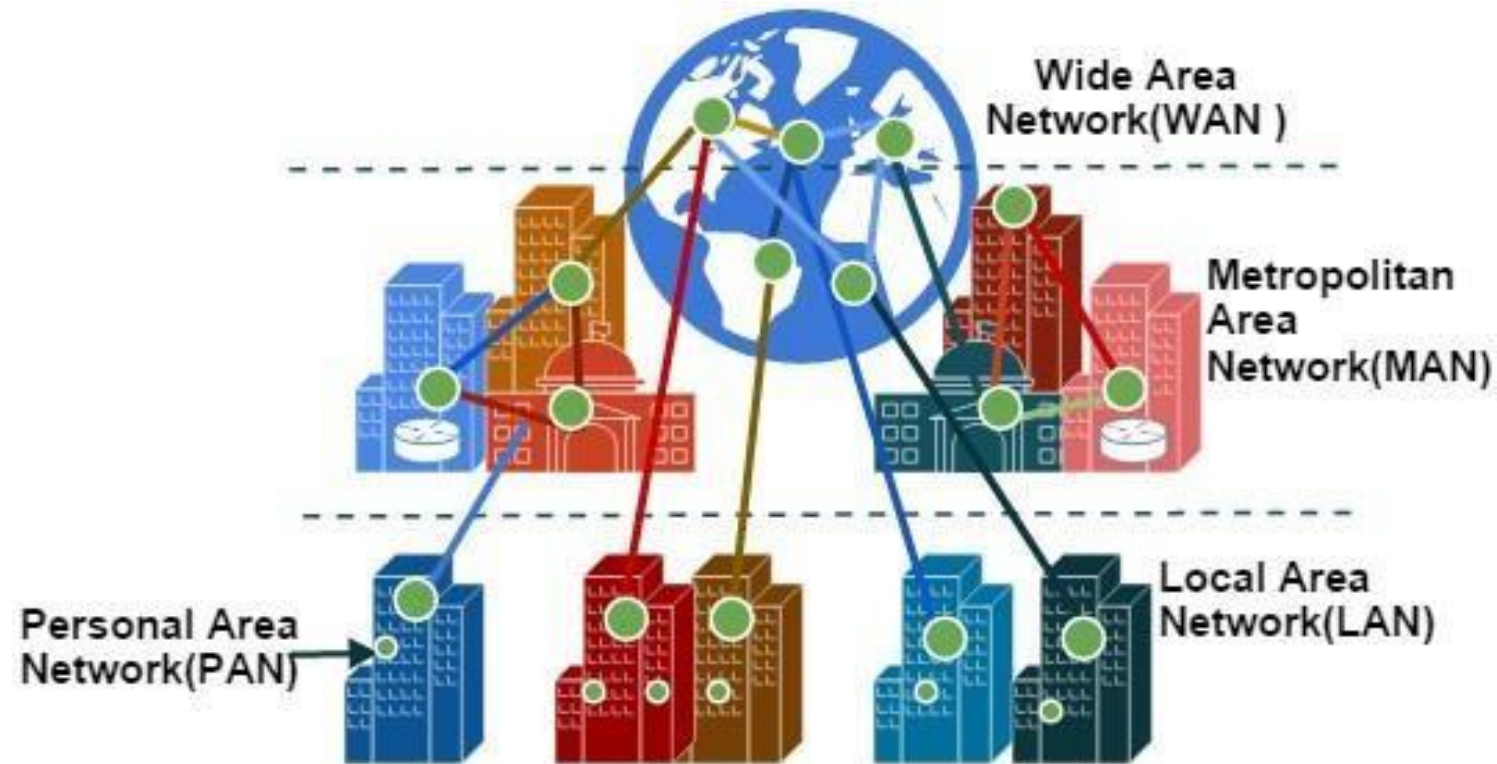
Windows IP Configuration

Ethernet adapter Local Area Connection:

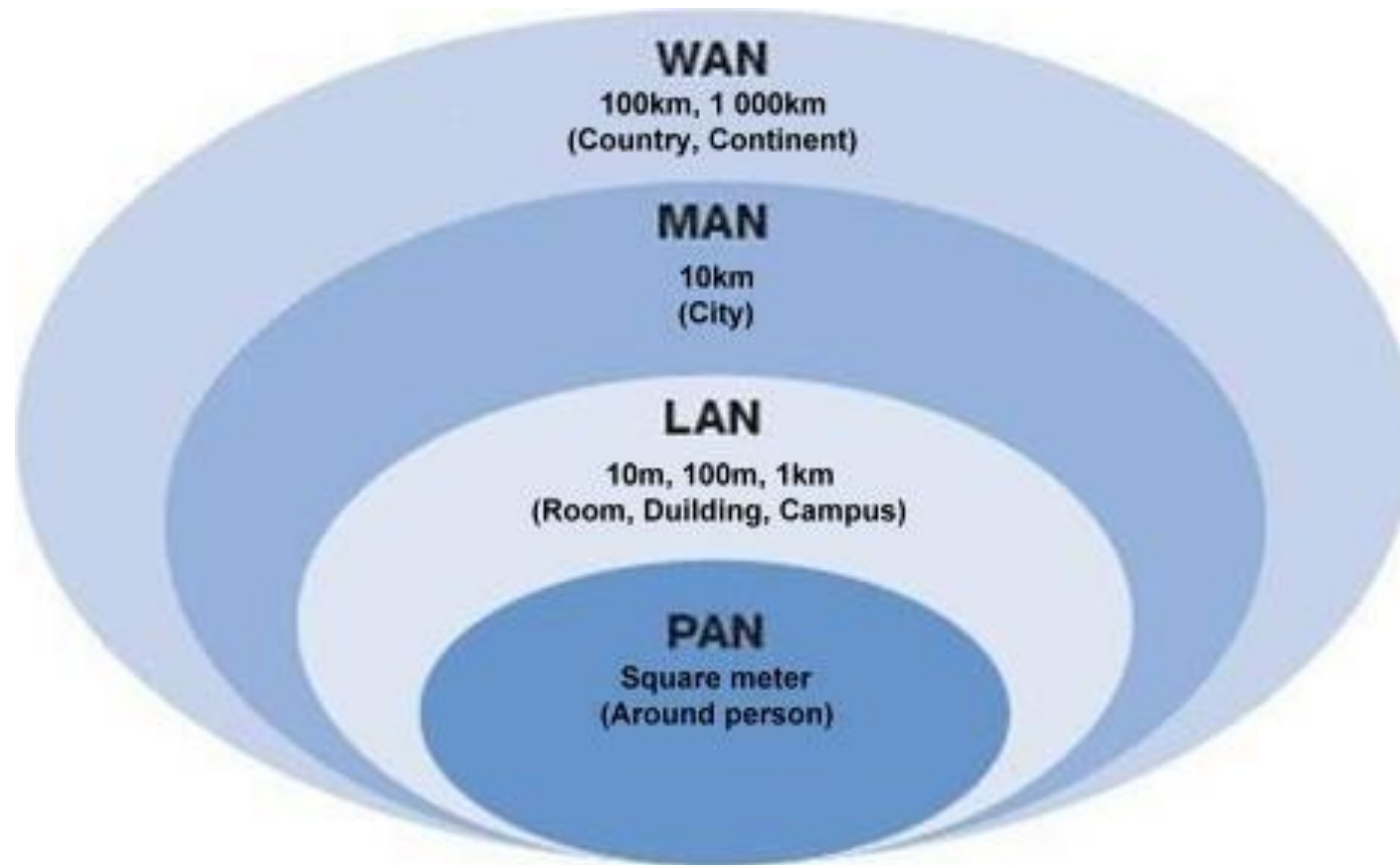
    Connection-specific DNS Suffix  . : hst.wa.comcast.net.
    Link-local IPv6 Address . . . . . : fe80::dcd8:42af:
    IPv4 Address. . . . . : 192.168.1.125
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
```

You can check IP address by typing **ipconfig** command using command prompt

Types of Networks

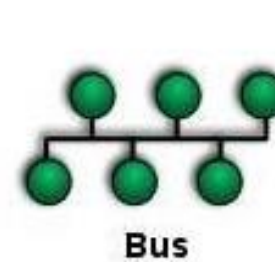
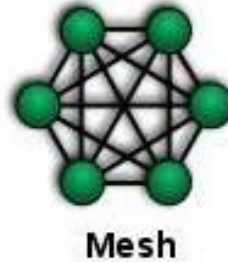
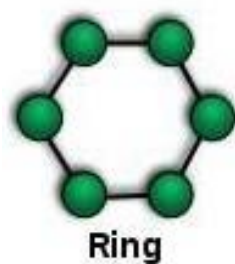


Types of Networks



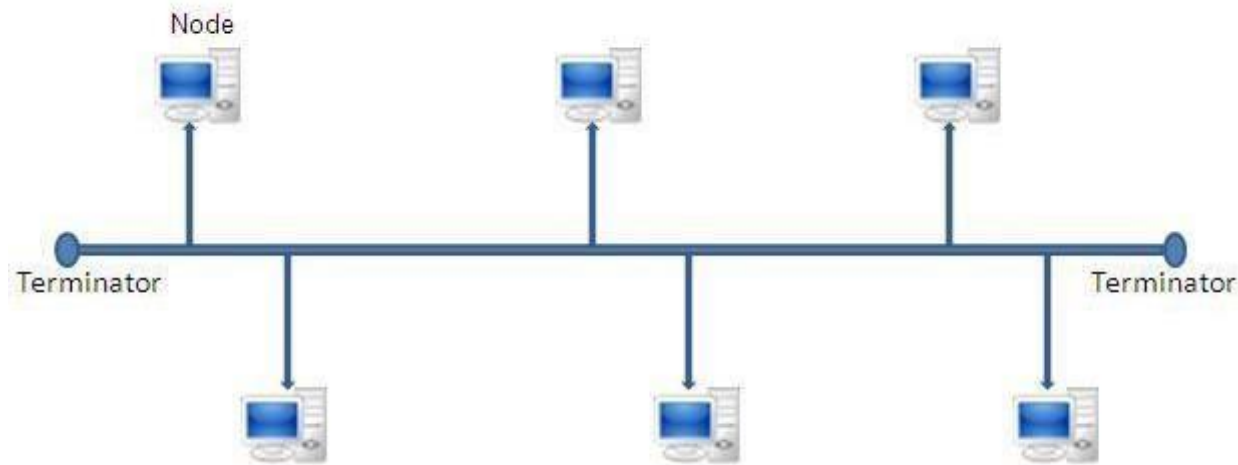
Network Topologies

- ❑ A network topology is the pattern in which nodes (computers, printers, routers or other devices) are connected to a local area network (LAN) or other network via links (E.g. twisted pair, copper wire cable or optical fiber cable)



Bus Topology

- ❑ Is a network type in which every computer and network device is connected to single cable.

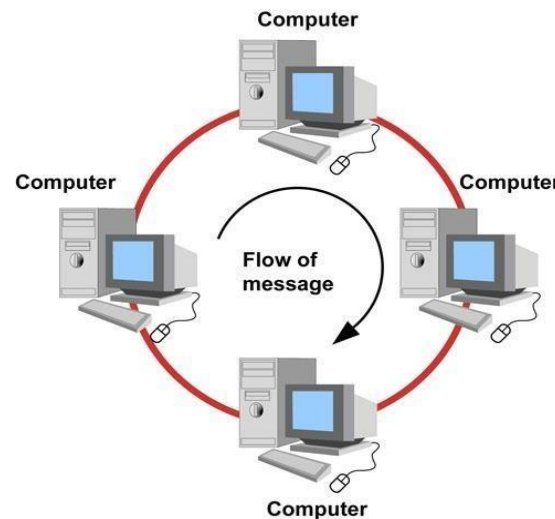


Bus Topology

Advantages	Disadvantage
Cost effective	Cable fails then whole network fails
Cable required is least compared to other network topologies	If network traffic is heavy or nodes are more the performance of the network decreases.
Used in small networks	Cable has a limited length
It is easy to understand	Slower than the ring topology.

Ring Topology

- ❑ The nodes in the network are connected in a circular fashion, and the data travels in one direction.
- ❑ Exactly two neighbors for each device is available.
- ❑ Sending and receiving of data takes place by the help of Token.



Ring Topology

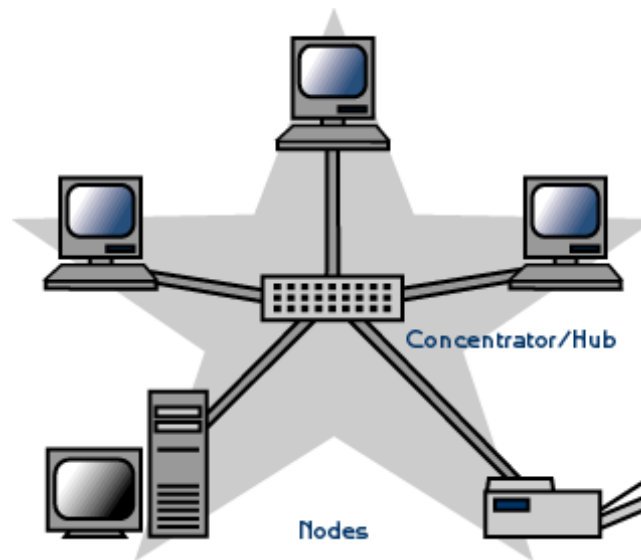
- ❑ **Token passing** : A token is regularly passed from one node to another. And if a node has some information to pass on the network, the node releases the information. If the node does not have any data to release on the network, then it transfers the token to the next node.
- ❑ The nodes with token are the ones only allowed to send data. Other nodes have to wait for an empty token to reach them.

Ring Topology

Advantages	Disadvantage
Reduced chances of data collision as each node release a data packet after receiving the token	Each packet of data must pass through all the nodes between source to destination
Token passing makes ring topology perform better than bus topology under heavy traffic	If one node goes down, the entire network gets affected
No need of server to control connectivity among the nodes	
Equal access to the resources	

Star Topology

- ❑ Each computer is connected to a central hub using point to point connection.
- ❑ The central hub can be a simpler device that only makes the connections between computers over the network.

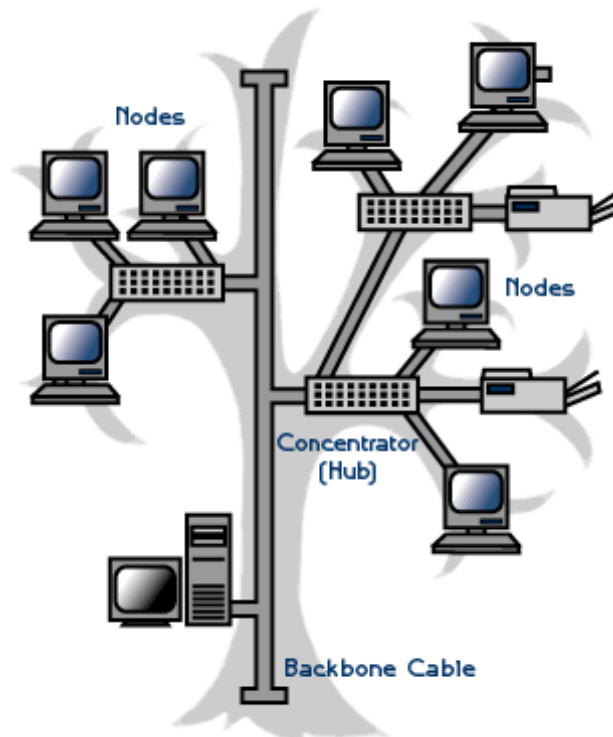


Star Topology

Advantages	Disadvantage
As compared to Bus topology much better performance, signals don't necessarily get transmitted to all the node.	Too much dependency on central device. If it fails whole network goes down.
Easy to connect new nodes or remove without affecting rest of the network.	The use of hub, router or switch as central device increase the overall cost of the network.
Failure of one node or link doesn't affect the rest of network.	Performance and the number of nodes which can be added is depend on capacity of central device.

Tree Topology

- ❑ Combines multiple star topologies on to a bus.
- ❑ Hub device for each star topology are connected to the bus.
- ❑ This is also known as Expanded star topology.

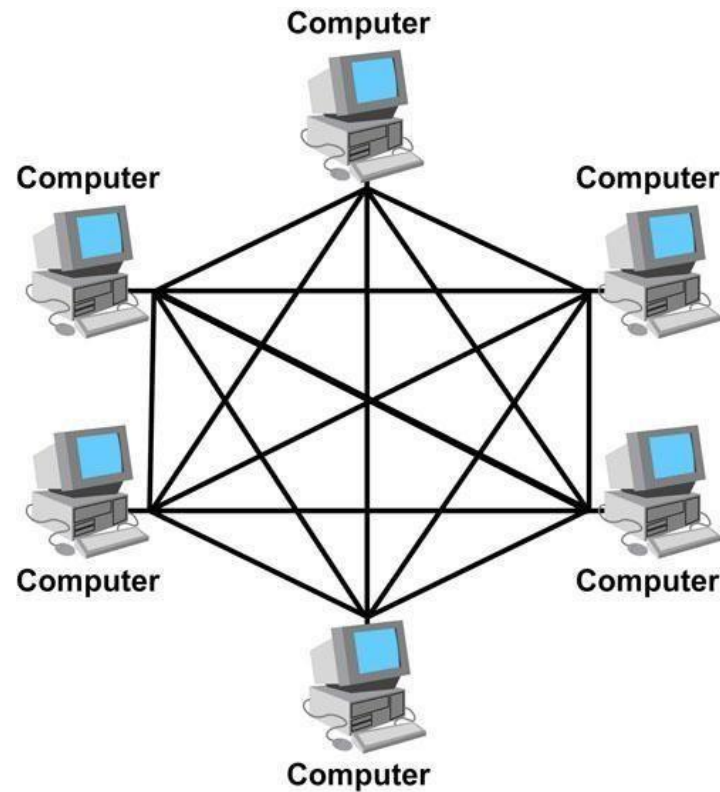


Tree Topology

Advantages	Disadvantage
Expansion of network is possible and easy.	Relies heavily on the main bus cable, if it breaks whole network is crippled.
Here, we divide the whole network into segments (star network) which can be easily managed and maintained.	Scalability of the network depends on the type of cable used.
Error detection and correction is easy.	
If one segment is damaged, other segments are not affected.	

Mesh Topology

- ❑ Every node has a direct point to point connection to every other node.

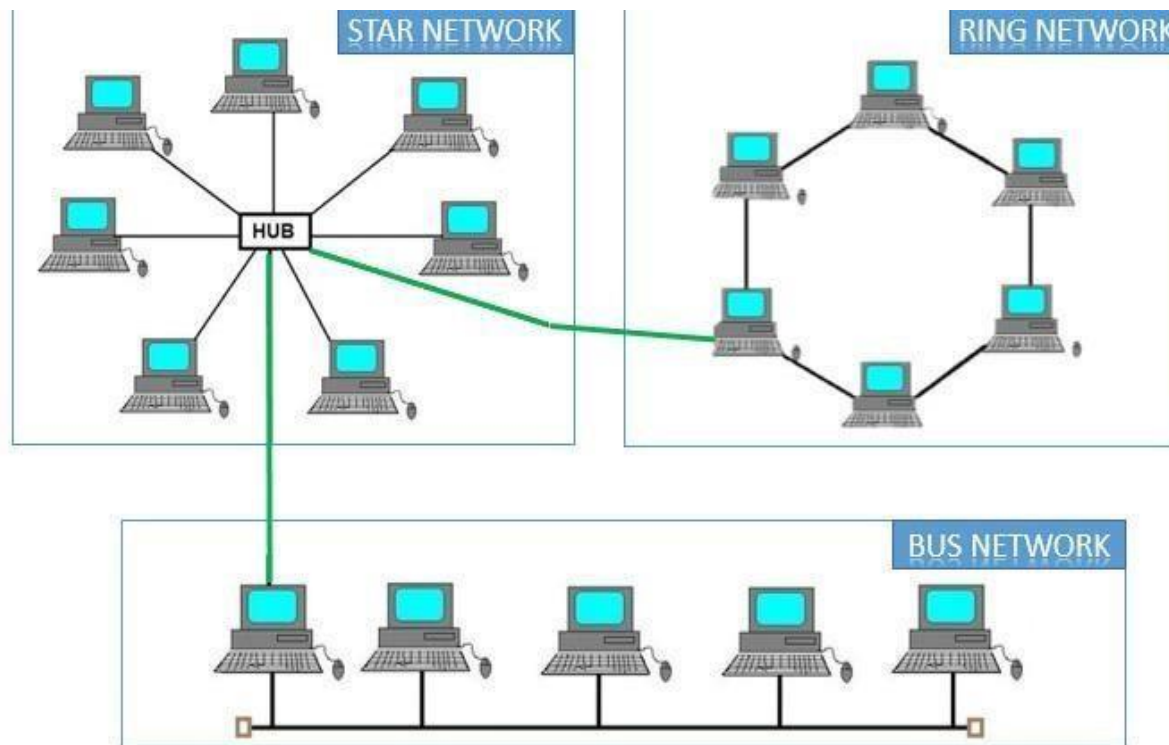


Mesh Topology

Advantages	Disadvantage
Even if one of the components fails there is always an alternative present. So data transfer doesn't get affected.	Requires a lot of cable, therefore it is expensive.
Because all connections are direct, the network can handle very high- volume traffic.	Many of the connections are also redundant since there are several different paths for data to travel from one node to another.
Security is high because data travels along a dedicated connection.	Set-up and maintenance of this topology is very difficult.

Hybrid Topology

- ❑ Uses a combination of two or more topologies.
- ❑ Tree topology is one specific example of a star bus network.



Hybrid Topology

- ❑ Home Work
 - ❑ Find out advantages and disadvantages of the Hybrid topology.

What are Networking devices

- ❑ Network devices are components used to connect computers or other electronic devices together so that they can share files or resources like printers or fax machines.
- ❑ Devices used to setup a LAN are most common type of network devices used by public.

Examples

- ❑ NIC (Network Interface Card)
- ❑ Repeater
- ❑ Modem
- ❑ Bridge
- ❑ Hub
- ❑ Switch
- ❑ Router

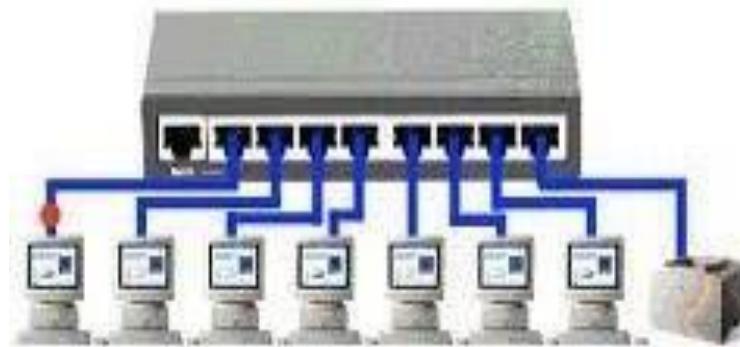
NIC (Network Interface Card)

- ❑ Is a computer hardware component that connects a computer to a computer network.
- ❑ Also known as network interface controller, network adapter, LAN adapter or physical network interface.
- ❑ In most of the newer computers have a network interface built into the mother board.



Hub

- ❑ Is a most basic networking device that connects multiple computers or other network devices together.
- ❑ A hub as no routing tables or intelligence on where to send information and broadcast all network data across each connection.



Switch

- ❑ Is a hardware device that connects devices together on a computer network and it forwards data only to one or multiple devices that need to receive it.



Hub



Switch

Repeater

- ❑ Is an electronic device that receives a signal and retransmits it.
- ❑ These are used to extend transmission so that the signal can cover longer distance.



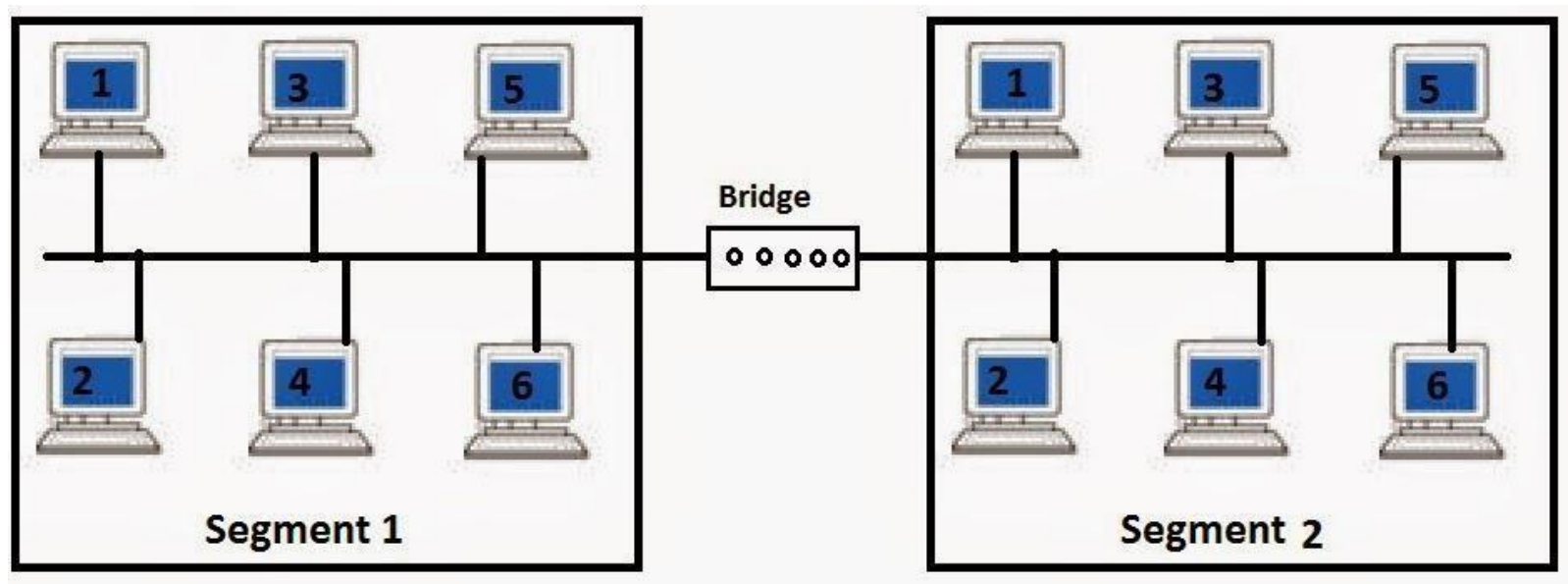
Router

- ❑ A networking device that forwards data packets between computer networks.
- ❑ The router will also choose the best route for the data packet so that destination node can receive the information quickly.



Bridge

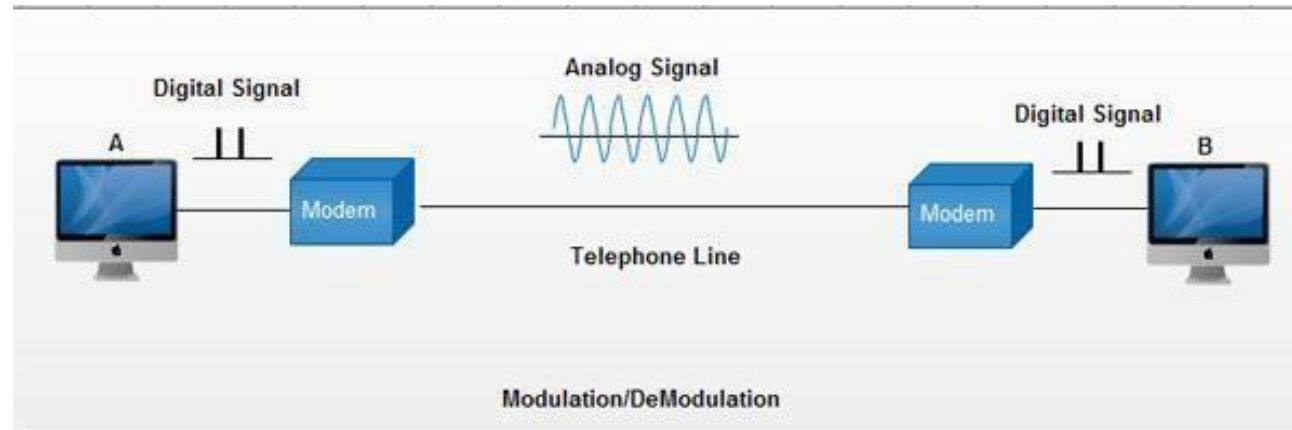
- ❑ A device that connects a local area network (LAN) to another are network that uses the same protocol



Modem

- ❑ Modem stands for **MOD**ulator **dEM**odulator, is a device that enables a computer to transmit data over telephone lines.
- ❑ Computer information is stored digitally, whereas information transmitted over telephone line is use analog waves.
- ❑ A modem converts between these two forms.
 - Modulator : Convert information from digital mode to analog mode at the transmitting end.
 - Demodulator: Convert information from analog to digital at receiving end.

Modem



THANK YOU